

Medical Robotics

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Lecture 7

Control Modalities of Medical Robots

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1. Context: Control Modalities in the Reference Design Table

The course analyses medical robotic systems column by column through the reference design table. With the kinematic design module now complete, the course proceeds to the second column of the reference design table: the control modality. The kinematic architecture tells us *what* the robot can do (its reachable workspace, its mobility, and its singularities); the control modality tells us *how* the robot's motion is commanded — and in particular, *who (or what)* is in the control loop.

Domain use	of Robotic task	Kinematic architecture	Control modality	Sensors & actuators
Orthopedics	Machining of rigid surface	Conventional serial (SCARA, anthropomorphic, spherical); 5 or 6 DOF	Autonomous; cooperative or “hands-on”	Conventional; vel: mm/s; force: up to 100 N
MIS	Constrained manipulation	Passive joints; mechanical RCM (parallelogram, spherical, ...); 5 or 6 external + extra internal DOF	Teleoperation; shared control	Conventional; vel: cm/s; force: few N
Neurosurgery, int. radiology, radiotherapy	Constrained targeting	Conventional + front-end with dedicated architecture; 5 or 6 DOF	Semi-autonomous; teleoperation; autonomous	MR/CT compatible (pneumatic, piezo, ultrasonic); force: few N
Microsurgery	Micromanipulation	Dedicated kinematic architecture	Shared/cooperative; teleoperation	Piezo, ultrasonic; force: few mN; vel: 0.70 m/s
Tele-echo, TMS, skin harvesting	Surface tracking	Conventional + dedicated wrist architecture	Autonomous; teleoperation; shared	Conventional; vel: mm/s; force: few N

Table 1: Reference design table; control modality column highlighted.

2. The Three Control Modalities

Three main control modalities.

- **Autonomous(task-level autonomy):** the robot executes a pre-planned surgical task with no real-time human input beyond monitoring and emergency intervention. Example: ROBODOC autonomously mills the femoral canal during hip replacement.

The robot is in contact with the patient who is expected not to move except for involuntary physiological motion. The central control problem is *interaction with a non-cooperating patient*. The key design requirement is safe physical interaction: the robot must never injure the patient, yet it must apply force to perform the task (drilling, cutting, ablating). This apparent contradiction — a robot that must apply force but must never cause unintended harm — is the central challenge of autonomous medical

robot control.

- **Cooperative / hands-on / shared control:** the robot constrains or augments the surgeon's motion in real time. The surgeon physically moves the robot, and the robot either filters the motion, applies virtual fixtures, or prevents entry into forbidden zones. Example: the ACROBOT/MAKO system, which confines the surgeon's motion to a safe cutting volume.

The robot is in simultaneous contact with both the patient and the surgeon. The central control problems are *interaction with the non-cooperating patient* and *cooperation with the surgeon*. The control law must manage interaction with both, under fundamentally different conditions. The surgical instrument force exerted on the patient

- **Teleoperation:** the remotely operated slave robot is in contact with the patient; the master, guided by the surgeon with possible force feedback, is a separate robot. If the master is a passive haptic device (an articulated device with no active force feedback), the surgeon receives no force information; if it is an active haptic master, the interaction forces at the slave are scaled and reflected to the surgeon, creating a bilateral teleoperation loop. Example: the da Vinci surgical system for minimally invasive surgery.

The central control problems are *interaction with the non-cooperating patient* (slave side), *interaction with the surgeon* (master side), and the *control of the remote slave*, possibly shared between the surgeon and a local controller.

Modality vs. control law. The control modality describes the role of the human in the control loop; it does not prescribe a specific control law. The same control law (e.g. impedance control) can appear in all three modalities. Conversely, the modality determines which *control problems* must be solved: a cooperative system must manage human-robot interaction on both the patient and surgeon sides simultaneously, a constraint that does not arise in purely autonomous operation. The following sections begin to bridge from modality to law.

2.1. Common Control Problem: Physical Interaction

All three modalities share one fundamental control problem: **managing the physical interaction between the robot and a human** (patient, surgeon, or both). The control objectives are to regulate the interaction forces and to ensure accurate execution of the surgical plan. The methodologies available are:

- **Impedance control** (with or without force measurement): controls interaction force through motion.
- **Admittance control** (with force measurement or estimation): controls interaction force through motion; force measurement is required.
- **Force control and hybrid position/force control:** explicit control of interaction force; require force measurement.
- **Active constraints / virtual fixtures** (impedance/admittance type): used in cooperative and shared-control modalities.

This lecture focuses on impedance and admittance control, with the active constraints paradigm developed through the Acrobot case study.

3. Problem Abstraction: Robot Dynamics in Contact

3.1. The Cartesian Dynamic Model After Feedback Linearisation

A rigorous treatment of robot dynamics uses the full inertia, Coriolis, and gravity model. After feedback linearisation in Cartesian coordinates (covered in full detail in the Robotics II course), the closed-loop dynamics of a robot in contact with the environment can be written as:

$$M_p(\mathbf{q}) \ddot{\mathbf{p}} = M_p(\mathbf{q}) \mathbf{a} + \mathbf{F}_{\text{ext}}$$

where $\mathbf{p} = (x, y)^T$ (or $(x, y, z)^T$ in 3D) is the Cartesian position of the end-effector, $M_p(\mathbf{q})$ is the Cartesian inertia matrix (a function of the joint configuration $\mathbf{q}$), $\mathbf{a}$ is the acceleration input (the control variable after linearisation), and $\mathbf{F}_{\text{ext}}$ is the external force arising from contact with the environment.

3.2. Simplification to a Mass in Cartesian Space

Assuming $M_p(\mathbf{q})$ is diagonal and constant with all entries equal to a scalar mass m , the model reduces to:

$$m \ddot{\mathbf{p}} = \mathbf{u} + \mathbf{F}_{\text{ext}}$$

where $\mathbf{u} = m\mathbf{a}$ is the control force. This is precisely Newton's second law for a point mass m subject to a control force $\mathbf{u}$ and a contact force $\mathbf{F}_{\text{ext}}$.

Why this simplification is useful. The mass model concentrates all of the design insight in a one- or two-dimensional scalar problem. The geometry and computation needed for a real 3D manipulator are well-established (inverse kinematics, Jacobian-based force mapping); once the interaction model is designed at the task level, it is mapped to joint torques by standard techniques. The simplification does not sacrifice the essential physics of the interaction control problem.

4. Impedance Control

4.1. The Impedance Interaction Model

Impedance model. The *input* is the motion (position, velocity, acceleration of the end-effector) and the *output* is the interaction force. The robot “impedes” the motion by generating a reaction force. Desired interaction:

$$F_{\text{ext}} = m_d(\ddot{p} - \ddot{p}_d) + b(\dot{p} - \dot{p}_d) + k(p - p_d) \quad (*)$$

where $p_d(t)$ is the desired task trajectory, m_d is the desired apparent mass, b the desired damping, and k the desired stiffness.

The three terms of the impedance model have clear physical interpretations:

- The **stiffness term** $k(p - p_d)$ acts like a spring attracting the end-effector back to the desired trajectory. If p_d is set to the current position (no independently planned trajectory), this term vanishes and the robot offers no positional resistance.
- The **damping term** $b(\dot{p} - \dot{p}_d)$ resists velocity deviations from the desired motion, providing a viscous-like resistance.
- The **inertia term** $m_d(\ddot{p} - \ddot{p}_d)$ shapes how easily the robot can be accelerated by an external force; by choosing $m_d < m$, the robot appears *lighter* than it physically is, assisting the operator.

4.2. The Impedance Control Law

To enforce (*), choose the acceleration input as:

$$a = \ddot{p}_d + \frac{1}{m_d} [b(\dot{p}_d - \dot{p}) + k(p_d - p) + F_{\text{ext}}]$$

which, after substituting $\mathbf{u} = m\mathbf{a}$ into $m\ddot{\mathbf{p}} = \mathbf{u} + F_{\text{ext}}$, yields the control law:

$$u = m\ddot{p}_d + \frac{m}{m_d} [b(\dot{p}_d - \dot{p}) + k(p_d - p)] + \left(\frac{m}{m_d} - 1 \right) F_{\text{ext}}$$

4.2.1. Special case: no modification of apparent inertia ($m_d = m$)

When the apparent mass equals the actual mass, the last term vanishes:

$$u = m\ddot{p}_d + b(\dot{p}_d - \dot{p}) + k(p_d - p)$$

This is a **PD control law with a feedforward acceleration term**. It requires *no force measurement*. The parameters b and k play the roles of the derivative and proportional gains, and they directly shape the perceived interaction dynamics.

4.2.2. Case: modified apparent inertia ($m_d \neq m$)

The last term involves F_{ext} , requiring a force/torque sensor at the end-effector. The ratio m/m_d amplifies (or attenuates) the measured force contribution, allowing the designer to make the robot feel lighter or heavier than it physically is.

Sensor requirement summary. Shaping the stiffness k and damping b of the interaction *does not require force sensing* — only position and velocity feedback (from encoders) is needed. Adding force sensing is only necessary when the designer also needs to reshape the apparent inertia. This has significant practical consequences: force/torque sensors add cost, weight, calibration burden, and a potential failure point. Operating with $m_d = m$ avoids this dependency.

5. Admittance Control

5.1. The Admittance Interaction Model

Admittance model. The *input* is the force applied to the robot and the *output* is the admitted motion. The robot generates a reference motion p_r in reaction to the contact force so as to enforce:

$$F_{\text{ext}} = m(\ddot{p}_r - \ddot{p}_d) + b(\dot{p}_r - \dot{p}_d) + k(p_r - p_d)$$

In the Laplace domain, defining $\Delta p_r = p_r - p_d$:

$$\Delta p_r = \frac{F_{\text{ext}}}{ms^2 + bs + k}$$

The key distinction from impedance control is that p_r is *not* the current position of the robot but a *reference motion generated by the model* as a consequence of the applied force. This reference is then fed to an inner position controller.

Impedance vs. admittance: the structural difference. In impedance control, the control law u directly commands joint torques (or an equivalent Cartesian force). In admittance control, the outer admittance model computes a modified reference position $p_r = p_d + \Delta p_r$, which is then tracked by an inner position controller (e.g. a PD controller). Admittance

control is therefore well suited to robots with stiff, non-backdrivable actuators and good position control, since the inner controller can reliably track the reference. Impedance control is more natural for intrinsically backdrivable, low-inertia robots. **Force measurement is always required in admittance control**, since F_{ext} is the input to the admittance model.

6. Application Example 1: Impedance and Admittance Control for Superficial Hyperthermia (ROBHOT / Alba Robot)

6.1. Clinical Background

Superficial hyperthermia is a cancer treatment used as an adjuvant to radiotherapy (and sometimes chemotherapy). It consists of raising the temperature of tumour tissue by 4–8 °C above the physiological baseline, typically by means of an electromagnetic antenna applied externally. The mechanisms by which hyperthermia enhances radiotherapy efficacy include: inhibition of DNA repair in tumour cells after radiation damage (the cellular self-repair process that normally reconstructs damaged DNA is suppressed at elevated temperatures), increased absorption of chemotherapy agents, and improved tumour oxygenation.

Superficial hyperthermia is applicable when the tumour target lies within the penetration depth of the radiated field (typically a few centimetres), making it particularly relevant for breast cancer, skin cancer, and head-and-neck tumours.

Why robotic assistance? Until recently, hyperthermia treatments were delivered by hand using a passive arm with friction-locked joints, with no quantitative registration between the pre-treatment plan and the actual antenna position. The effectiveness of hyperthermia is highly sensitive to the delivered dose, which in turn depends on the antenna pose relative to the tumour target. Robotic assistance was introduced to: (1) accurately transfer the pre-treatment plan to the robot; (2) maintain the planned antenna pose during the treatment (which lasts 40–50 minutes); (3) allow the robot to be compliant with patient respiratory motion; and (4) enable manual guidance by the clinical staff during initial positioning.

6.2. The ROBHOT / Alba Robot System

The system (developed in collaboration between Sapienza University of Rome, Medlogix s.r.l., and the ALBA ON4000 platform) consists of a serial robot arm carrying a water-bolus-cooled electromagnetic antenna. A pre-treatment planning system computes the desired antenna pose in a 3D patient model; this plan is transferred to the robot via a registration step that aligns the imaging coordinate frame, the robot base frame, the patient frame, and the camera frame in the operating room.

The control architecture implements two distinct interaction control tasks:

1. **Autonomous antenna tracking** (robot–patient interaction): impedance control maintains contact between the antenna and the patient’s skin surface with a prescribed contact force, while the robot tracks respiratory motion.
2. **Manual guidance** (robot–operator interaction): admittance control allows the clinical staff to manually position the robot in space by applying forces to the arm, making the robot feel light and compliant.

6.3. Impedance Control for Antenna Tracking

The abstract problem is modelled exactly as in Section 4: the antenna is treated as a mass m whose task is to reach a desired target position (slightly inside the patient model, to ensure

maintained contact) and then track the patient's breathing motion.

The interaction force in the contact direction (normal to the skin surface, approximately the z -direction in the task frame) follows the impedance model. The parameters chosen were: $M_m = M$ (no inertia reshaping, therefore no force sensor required for this phase), $K_m = 30$, $D_m = 15$, desired approach velocity $v_{\text{des}} = 0.005$ m/s. These parameters were determined by the clinical requirement that the steady-state contact force remain within approximately 2 N in the normal direction.

The key behaviours observed in simulation and experiment:

- During the approach phase, the antenna moves toward the target with a prescribed velocity profile.
- Upon contact, the impedance model becomes active: the robot does not simply press against the skin with increasing force, but instead reaches a force equilibrium determined by k (the stiffness) and the position error in the normal direction.
- During patient breathing, the robot tracks the skin surface motion while maintaining the target contact force within a bounded range.
- Position errors in the x - and y -directions (tangential to the skin) remain near zero; the error in z (normal direction) is a few centimetres, which is acceptable because it is the *force* in z that is being controlled, not the position.

Force vs. position control in the contact direction. This example illustrates a fundamental principle of interaction control: in the direction of contact, it is often inappropriate to control position (the robot would press harder and harder against the patient). Instead, one controls the force in the contact direction (via impedance) and the position in the tangential directions. This is the essence of hybrid position/force control, of which impedance control is a soft-contact approximation.

6.4. Admittance Control for Manual Guidance

Manual guidance is implemented as admittance control: the operator applies a force to the robot arm (detected by a wrist force/torque sensor), and the admittance model generates a reference velocity for the robot proportional to the applied force. The robot then moves in the direction of the applied force, appearing to have a very low inertia regardless of its physical mass.

The admittance model is:

$$V(s) = Y(s) F(s)$$

where $F(s)$ is the measured interaction force (robot–operator), $Y(s)$ is the device admittance (design parameter), and $V(s)$ is the reference velocity. By choosing $Y(s)$ appropriately (e.g. a pure scalar gain or a first-order filter), the responsiveness and smoothness of the manual guidance can be tuned.

7. Cooperative / Hands-On Systems

7.1. Definition and Properties

In a cooperative (hands-on) system, both the robot and the surgeon physically interact with the same surgical instrument simultaneously. The control scheme detects the direction in which the surgeon wishes to move the tool (through a force sensor at the handle) and commands the robot to comply, subject to pre-defined spatial constraints.

Key properties of this control paradigm:

- Combines robotic precision, strength, and tremor suppression with the immediacy of freehand surgical manipulation.
- Surgeons find it intuitive and natural; it preserves the feeling of being in control.
- Less expensive than full teleoperation (no second robot, no communication infrastructure).
- **Force scaling is allowed** (the robot can amplify or attenuate the guiding force); **position scaling is inherently not permitted** (the surgeon's hand and the tool tip move by the same amount).
- **Incompatible with remoteness**: the surgeon must be physically co-located with the tool; instruments requiring distal dexterity (flexible tools, tools inside a body cavity) cannot be controlled this way.
- Typical domains: orthopaedic surgery, microsurgery.

7.2. Actuator Requirements and Admittance Type

Cooperative systems typically use **non-backdrivable, velocity-controlled actuators**. Non-backdrivability is essential: a backdrivable actuator would comply with any external force, including unintended disturbances from the patient or the environment, not just the intentional guiding force of the surgeon. By being non-backdrivable, the robot resists all external forces except those explicitly commanded through the admittance law.

The control model is the linear admittance:

$$V(s) = Y(s) F(s)$$

where $F(s)$ is the measured surgeon–robot interaction force, $Y(s)$ is the device admittance, and $V(s)$ is the desired velocity. The admittance $Y(s)$ encodes the responsiveness of the system and its choice is constrained by stability: if the admittance provides highly prompt response, the closed-loop system can become unstable, particularly when the tool is in contact with stiff tissue.

Smoothness can be improved by thresholding and filtering the command force to suppress involuntary tremor. Haptic feedback and virtual fixtures can also be incorporated into the admittance model.

8. Application Example 2: The Acrobot / MAKO Active Constraints Robot

8.1. System Overview and Clinical Motivation

The Acrobot (Active Constraints Robot) was developed at Imperial College London for **bone machining in knee replacement surgery**. Its descendant, the MAKO system (acquired by Stryker in 2013), is today widely deployed in clinical practice for both partial and total knee replacement.

The clinical workflow:

1. **Pre-operative planning**: a CT scan of the patient's knee generates a 3D model. The size, position, and orientation of the prosthetic implant are planned patient-specifically. Cutting constraints (the volume of bone to be removed and the surfaces to be preserved) are computed and stored.
2. **Intra-operative registration**: the 3D model is aligned with the physical patient in the operating room using bone registration landmarks.

3. **Dynamic joint balancing:** intra-operative kinematic assessment allows the surgeon to modify the pre-operative plan to optimise ligament balance.
4. **Robotic-arm-assisted bone preparation:** the surgeon guides the cutting tool by hand, while the robot enforces the pre-planned spatial constraints using active constraints control.

8.2. Robot Hardware

The original Acrobot end-effector (2003) was a spherical manipulator with three orthogonal axes: yaw, pitch (within $[-30, 30]$), and extension (30–50 cm). Key design choices:

- **Low mechanical impedance** of all axes, so the surgeon can move the tool with minimal force within the permitted zone.
- A **six-axis force/torque sensor** in the handle to measure the surgeon's guiding force F_G .
- A **detachable, autoclave-sterilisable cutter motor** at the tip.
- Placement on a **six-axis gross positioning device** to extend the small working envelope of the spherical wrist to cover the full extent of the knee.

8.3. The Active Constraints Concept and Spatial Regions

The intervention space is partitioned into three regions based on the distance d from the tool tip X to the nearest point X_{np} on the planned boundary:

$$d = \|X - X_{np}\| = (X - X_{np}) \cdot N_{np}$$

where N_{np} is the outward normal to the boundary at X_{np} .

Three spatial regions.

- **RI** ($d > D_1$): the safe interior. The robot is fully compliant; the surgeon can move the tool freely in all directions. The robot's mechanical impedance is low and should be imperceptible.
- **RII** ($0 < d \leq D_1$): the transition zone. The robot's stiffness increases progressively as the boundary is approached. Motion components directed toward the boundary are resisted; tangential motion and motion away from the boundary remain free.
- **RIII** ($d \leq 0$): the forbidden zone. The robot is stiff; the tool is prevented from further penetration. The only permitted motions are tangential to or away from the boundary.

The gradual stiffness increase in RII is essential to avoid instability: a sudden jump in stiffness at the boundary, combined with finite motor power, would cause unsafe oscillations. Abrupt boundary enforcement also cannot be guaranteed because the power of the actuators is always limited; a surgeon pushing hard can overshoot the boundary. The progressive stiffening in RII substantially reduces the overshoot error.

The boundary itself is a **2.5D constraint**: a flat plane (the cutting surface) combined with a 2D closed curve (the outline of the bone section to be cut), producing a slab-shaped region. The two parts of the boundary are orthogonal, so X_d , $\dot{X}_d$, and F_C can be computed separately for each part and combined before being passed to the inner loop.

8.4. Control Architecture

The controller has two nested loops running at different rates:

8.4.1. Inner loop: joint position/velocity control (2 ms, 550 Hz)

$$\tau = K_P(\Theta_d - \Theta) + K_D(\dot{\Theta}_d - \dot{\Theta}) + \tau_C + f^*(\Theta, \dot{\Theta}) + g^*(\Theta)$$

where τ is the torque vector commanded to the robot motors; K_P , K_D are the proportional and derivative gain matrices; Θ_d , $\dot{\Theta}_d$ are the desired joint position and velocity (computed by the outer loop); Θ , $\dot{\Theta}$ are the measured joint position and velocity; $\tau_C = J^T F_C$ is the torque compensation for the surgeon's guiding force (where J is the manipulator Jacobian); and f^* , g^* are friction and gravity compensation terms.

8.4.2. Outer loop: Cartesian admittance control (10–15 ms, 100 Hz)

The outer loop operates in Cartesian space (where the task constraints are defined) and computes the desired position X_d and velocity $\dot{X}_d$ (converted to joint space via inverse kinematics: Θ_d , $\dot{\Theta}_d = J^{-1}\dot{X}_d$), the PD gains K_P and K_D , and the compensation force F_C . The computation depends on the current region RI, RII, or RIII:

Region RI ($d > D_1$): free motion.

- PD gains are small (low impedance) so the surgeon feels the cutting forces.
- $X_d = X$ (desired position equals current position; no restoring force).
- $\dot{X}_d = A F_G$ (admittance: velocity is proportional to guiding force, isotropic in all directions).
- $F_C = 0$ (no force compensation; the surgeon's guiding force is not opposed).

Region RII ($0 < d \leq D_1$): transition. The guiding force F_G is decomposed into normal and tangential components with respect to the boundary:

$$F_{GN} = (F_G \cdot N_{np}) N_{np}, \quad F_{GT} = F_G - F_{GN}$$

The desired velocity becomes:

$$\dot{X}_d = A F_{GT} + A_N F_{GN}, \quad A_N = \frac{d}{D_1} A$$

The normal admittance A_N decreases to zero as the tool approaches the boundary ($d \rightarrow 0$), progressively blocking normal motion. Tangential motion remains fully admitted. The compensation force is non-zero for $d \leq D_2 < D_1$:

$$F_C = -\frac{D_2 - d}{D_2} F_{GN} \quad (d \leq D_2), \quad F_C = 0 \quad (d > D_2)$$

Region RIII ($d \leq 0$): forbidden zone.

- PD gains are high (high stiffness, resisting penetration).
- $X_d = X_{np}$ (desired position is the nearest boundary point; the robot drives the tool back to the boundary).
- If F_G points away from the boundary, $\dot{X}_d$ and F_C are set as in RI (tool can escape freely).
- Otherwise: $\dot{X}_d = A F_{GT}$ and $F_C = -F_{GN}$ (tangential motion admitted, normal motion fully compensated).

Why the two-loop architecture? The inner loop must run fast (550 Hz) to achieve stable joint control and good disturbance rejection. The outer loop, which implements the admittance logic and computes desired positions and velocities, needs to be computationally heavier (Cartesian constraint evaluation, boundary distance computation) but does not need to match the inner loop bandwidth. The separation of time scales allows each loop to be designed and tuned independently, a standard approach in multi-rate control.

8.5. Clinical Outcomes and Commercialisation History

The Acrobot demonstrated 2–3 times better accuracy and 3 times better reproducibility in bone preparation compared to manual freehand surgery, with 92% patient satisfaction at two years. The technology transfer timeline:

- 1999: Acrobot Company founded after 8 years of research at Imperial College London.
- 2010: Acrobot Company acquired by Stanmore Implants Worldwide Ltd (SIW).
- 2013: SIW sold the Acrobot technology to the American company MAKO Surgical.
- Late 2013: Stryker Corporation purchased MAKO Surgical.

The system is now marketed as the **Stryker MAKO** platform for partial and total knee, as well as hip replacement surgery.

9. Summary and Key Conceptual Threads

- **All three control modalities share the problem of physical interaction control.** Whether autonomous, cooperative, or teleoperated, the robot is in contact with a human, and the contact force must be managed. Impedance and admittance control are the primary tools.
- **Impedance control does not require force sensing when the apparent inertia is not modified** ($m_d = m$). The resulting control law is a PD controller with a feedforward acceleration term. This is a significant practical advantage for surgical systems.
- **Admittance control always requires force measurement.** It generates a modified reference position or velocity for an inner position controller, making it well suited to robots with non-backdrivable, velocity-controlled actuators.
- **Active constraints (virtual fixtures) are a cooperative control strategy** implemented via spatially varying admittance. The key design idea is the progressive stiffening of the admittance as the tool approaches a pre-planned boundary, avoiding abrupt force discontinuities and instability.
- **The two-loop architecture** of the Acrobot (fast inner joint loop at 550 Hz; slower outer Cartesian admittance loop at 100 Hz) is a general pattern for cooperative medical robots where the constraint logic is computationally expensive but does not need to match joint control bandwidth.
- **Force decomposition into normal and tangential components** is the key geometric operation in active constraints: tangential motion is always admitted; normal motion is progressively blocked as the boundary is approached.
- **Hyperthermia as a clinical application** illustrates both impedance (robot–patient contact maintenance) and admittance (manual guidance by clinical staff) in a single system, and provides a clinically grounded example of how interaction control parameters are chosen from physical requirements (target contact force of ~ 2 N).

Key topics for revision.

1. The three control modalities, their task specificities, and the control problems each presents.
2. The simplified Cartesian mass model $m\ddot{p} = u + F_{\text{ext}}$ and the conditions under which it is valid.
3. The impedance interaction model (*): the role of m_d , b , k ; the control law; when force sensing is and is not required.
4. The admittance interaction model: the role of Δp_r ; the Laplace domain representation; the block diagram with inner position controller.
5. The structural difference between impedance and admittance control: direct torque command vs. modified reference for an inner loop; backdrivable vs. non-backdrivable actuators.
6. The ROBHOT/Alba system: impedance control for antenna tracking, admittance control for manual guidance, parameter selection.
7. The Acrobot/MAKO system: the three spatial regions (RI, RII, RIII), the distance metric d , the force decomposition into F_{GN} and F_{GT} , the control law in each region, and the two-loop architecture.
8. The 2.5D constraint geometry for bone cutting: plane boundary + 2D closed curve.

Bibliography

- [1] M. Jakopc, F. Rodriguez y Baena, S.J. Harris, P. Gomes, J. Cobb, B.L. Davies, “The hands-on orthopaedic robot ‘Acrobot’: early clinical trials of total knee replacement surgery,” *IEEE Trans. on Robotics and Automation*, vol. 19, no. 5, pp. 902–911, 2003.
- [2] G.D. Hager, A.M. Okamura, P. Kazanzides, L.L. Whitcomb, G. Fichtinger, R.H. Taylor, “Surgical and interventional robotics: Part III,” *IEEE Robotics and Automation Magazine*, vol. 15, no. 4, pp. 84–93, 2008.
- [3] N. Hogan, “Impedance control: An approach to manipulation,” *ASME Journal of Dynamic Systems, Measurement, and Control*, vol. 107, pp. 1–24, 1985.
- [4] R.J. Anderson and M.W. Spong, “Bilateral control of teleoperators with time delay,” *IEEE Trans. on Automatic Control*, vol. 34, no. 5, pp. 494–501, 1989.
- [5] M. Ferro, P. Pavoni and M. Vendittelli, “Robot-assisted superficial hyperthermia treatments: the ROBHOT system,” 2024 IEEE 20th International Conference on Automation Science and Engineering (CASE), Bari, Italy, 2024, pp. 1594–1599, doi: 10.1109/CASE59546.2024.10711644.